

CampusConnect

A Smart Campus Exchange Platform

Short Description

CampusConnect is a smart, AI-powered campus exchange platform that enables verified university students to buy, sell, rent, exchange, and give away items within a secure campus community. The platform leverages Artificial Intelligence to simplify listing creation, provide intelligent item recommendations, and automatically match wanted posts with newly listed items. By promoting the reuse of student resources, CampusConnect helps students save money, reduce waste, and foster a more sustainable and collaborative campus ecosystem.

Application Type

Type: Web Application

For Students (Client Side)

- Accessible from any device through a web browser without installation.
- Enables quick browsing, listing management, chatting, and transactions from anywhere on campus.

For Administrators (Admin Side)

- Provides a centralized dashboard to manage users, listings, disputes, reports, and analytics.
- Simplifies maintenance, updates, and monitoring without requiring changes on users' devices.

Motivation / Problem Statement

University students frequently require items such as textbooks, electronics, furniture, lab equipment, and other essentials for a limited period. At the same time, graduating or relocating students often possess items they no longer need but struggle to find suitable buyers or recipients. Existing online marketplaces are designed for the general public and lack campus-specific security, making transactions with unknown individuals risky and inconvenient.

Additionally, students often have difficulty locating specific items that are unavailable, resulting in unnecessary purchases of new products, increased expenses, and avoidable waste. There is also no centralized platform that facilitates buying, selling, renting, exchanging, and giving away items exclusively within a trusted university community.

CampusConnect addresses these challenges by providing a secure, AI-powered marketplace where only verified students can participate. The platform enhances the user experience through intelligent listing creation, automatic matching of wanted posts, real-time notifications, and secure transaction management, creating a safer, more efficient, and sustainable campus marketplace.

Target Users and Use Cases

Target Users

- University Students – Buy, sell, rent, exchange, or give away items, communicate with other students, manage transactions, and receive personalized recommendations and notifications.
- University Administrators – Moderate listings, resolve disputes, manage categories, analyze marketplace demand, and monitor overall platform activity.

Primary Use Cases — For Students

- Register and log in using a university email.
- Create, edit, and delete item listings.
- Buy, sell, rent, exchange, or give away items.
- Browse available items using categories and search.
- Contact sellers or buyers regarding listings.
- Manage purchases, rentals, and transaction history.
- Create wanted posts when required items are unavailable.
- Rate and review other users after successful transactions.
- Report inappropriate listings or users.

Primary Use Cases — For Administrators

- Approve or remove inappropriate listings.
- Manage item categories.
- Handle user reports and disputes.
- Monitor platform activity and usage.

Scope and High-Level Features

The scope of CampusConnect is to develop a web-based campus marketplace exclusively for verified university students. The platform focuses on enabling secure and efficient exchange of student-owned items while integrating Artificial Intelligence to improve usability and decision-making.

High-Level Features

- AI-assisted listing creation with automatic item recognition, category suggestions, condition detection, and estimated pricing.
- Intelligent matching between wanted posts and newly listed items.
- Real-time notifications for item matches, messages, listing updates, and rental reminders.
- Advanced filtering, saved searches, and personalized wishlists.
- Integrated in-app chat for secure communication between students.
- Complete rental management with request approval, due-date tracking, and return management.
- QR code/verification-based handover for secure item exchange.
- Separate payment and item receipt confirmation for improved transaction reliability.
- User reputation system with ratings and reviews.
- Personal sustainability dashboard showing money saved, items reused, and environmental impact.
- Semester-end reminders encouraging students to list unused items.
- Administrative analytics dashboard to identify high-demand items, marketplace trends, and user activity.
- OCR-based duplicate detection for notes and previous-year question papers.
- Audit logs and secure transaction records for accountability and traceability.

Technology Stack

CampusConnect will be built using the MERN stack (MongoDB, Express.js, React.js, and Node.js), chosen for its full JavaScript ecosystem, scalability, active community support, and suitability for building real-time, AI-integrated web applications.

Layer	Technology	Purpose
Frontend	React.js (Redux/Context API, React Router)	Responsive UI for browsing, listing, chat, dashboards
Backend (API)	Node.js with Express.js	REST API, auth, business logic, rental and transaction state
AI listing service	Python (FastAPI) + Groq API	Image recognition, category/condition detection, price estimation
Matching service	Node.js / Python + embeddings	Semantic matching between wanted-posts and listings, triggers match events
Messaging/event layer	Redis (Pub/Sub or Streams)	Backbone for match events, notification triggers, decoupling services
Scheduler	node-cron / BullMQ	Rental reminders, semester-end nudges, periodic analytics jobs
Database	MongoDB (Mongoose ODM)	Listings, users, transactions, chat history, reports
Real-time communication	Socket.io	In-app chat, live notification delivery to connected users
Notification delivery	Nodemailer (email), optional FCM (push)	Reach users for matches and reminders when not on the app
Authentication	JWT and bcrypt.js	University-email verified login and session management
OCR and duplicate detection	Tesseract.js / Google Vision + pHash / imghash	Text extraction plus near-duplicate detection for notes and PYQs
QR/code verification	qrcode (npm)	Secure item handover confirmation
Cloud storage	Cloudinary / AWS S3	Listing images, uploaded documents
Hosting and deployment	Vercel/Netlify (frontend), Render/AWS EC2/Heroku (backend), MongoDB Atlas	Deployment across all services
Version control	Git and GitHub	Source management, team collaboration

Roles and Responsibilities

The following team roles have been identified for the successful execution of the CampusConnect project. Responsibilities for each member are to be filled in below.

Role	Team Member	Enrollment No.	Responsibilities
Project Manager & System Architect	Dev Barot	202512069	Oversees project planning, system architecture, module integration, task coordination, and overall project execution. Owns cross-cutting platform infrastructure: centralized audit logging, global error-handling middleware, API rate-limiting, environment/secrets configuration, and system health monitoring.
Authentication & User Management Lead	Dhrumil Doshi	202512088	Develops secure authentication, user profiles, role-based access, and the ratings & reputation system. Owns payment-method integration, transaction receipt confirmation, and the personal sustainability dashboard (money saved, items reused, environmental impact).
Marketplace & Listing Management Lead	Nishka Shah	202512035	Defines item categories and listing workflows; builds listing CRUD APIs, category management, and image-upload backend. Develops the listing creation, browsing, and category UI; integrates AI-assisted listing suggestions with Heer Mehta; runs module-level testing on listing APIs and UI, then leads end-to-end testing of the buy, sell, rent, exchange, and giveaway workflows.
AI & Intelligent Services Lead	Heer Mehta	202512095	Researches AI and OCR requirements and evaluates Groq API feasibility; builds the FastAPI AI service scaffolding. Develops AI-assisted listing creation, price estimation, the semantic matching engine, and OCR-based duplicate detection for notes and question papers; integrates match-event triggers into the Redis pub/sub pipeline and benchmarks matching accuracy.
Search, Chat & Notification Lead	Heer Shah	202512024	Defines search, filtering, and notification requirements; builds search/filtering endpoints and the advanced search, filters, saved-search, and wishlist UI. Integrates match- and rental-reminder event triggers into the notification pipeline; implements Socket.io-based in-app chat and real-

Role	Team Member	Enrollment No.	Responsibilities
			time notifications, then tests chat reliability and delivery.
Rental & Transaction Management Lead	Charmi Bhayani	202512028	Defines rental and transaction requirements; develops rental request APIs and transaction state management. Integrates QR code generation and verification for secure handovers; builds the rental dashboard and due-date tracking UI; implements rental reminder scheduling feeding the shared notification pipeline; tests rental and transaction workflows end-to-end.
Admin Panel & Analytics Lead	Kajal Varlani	202512017	Defines admin requirements and moderation workflows; develops admin APIs for listing approval, user management, and dispute handling. Builds the admin dashboard and analytics views; coordinates with Deep Boghara on the analytics data pipeline; integrates reporting tools and moderation notifications, then tests admin and dispute-resolution workflows.
Database, DevOps & Quality Assurance Lead	Deep Boghara	202512076	Designs the database schema and sets up MongoDB Atlas, Git workflow, CI pipeline, and automated build checks. Supports backend/frontend integration and manages version control throughout; sets up the test-automation framework used for module-level checkpoints; monitors AI/real-time service performance, leads final unit, integration, and security testing, and manages deployment across Vercel/Netlify, Render/AWS, and MongoDB Atlas.

Timeline with Milestones

The project is planned across an estimated 14-week development cycle, divided into the following phases and milestones:

Phase	Milestone	Duration	Key Deliverables
Phase 1	Planning & Requirement Analysis	Week 1 - 3	Finalize scope, requirements gathering, wireframing, and technology setup.
Phase 2	UI/UX Design	Week 2 - 4	Design system, wireframes to high-fidelity mockups, and design review.
Phase 3	Core Backend Development	Week 3 - 10	Database schema, authentication, payment/receipt APIs, listing CRUD APIs, rental APIs, admin APIs, and audit-logging/infrastructure middleware.
Phase 4	Core Frontend Development	Week 5 - 9	React components for browsing, listing creation, payment & sustainability dashboards, rental dashboard, and admin views.
Phase 5	AI Feature Integration	Week 6 - 11	AI-assisted listing creation, price estimation, intelligent matching engine, and OCR duplicate detection.
Phase 6	Real-Time Features & Notifications	Week 9 - 12	In-app chat, real-time notifications, match/rental-reminder triggers via Socket.io and Redis pub/sub.
Phase 7	Continuous Module-Level Testing	Week 5 – 13	Rolling testing checkpoints after each module reaches feature-complete (auth, payment, listings, OCR, chat, rental, admin) — not held until the end.
Phase 8	Final Integration, Security & Regression Testing	Week 11 - 13	Cross-module integration testing, security checks, bug resolution, and full-system regression test.
Phase 9	Deployment & Final Presentation	Week 9 - 14	Performance monitoring, deployment to cloud (Vercel/Netlify, Render/AWS, MongoDB Atlas), documentation, and final project demo.

Project Timeline Chart (Gantt View)

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